Red-Agent Vulnerability Assessment

Automated Publication Bundle

April 9, 2026

Abstract

This report summarizes a full adversarial scan over the currently reachable model subset, combining campaign hit rates with StrongREJECT-style scoring and an explicit registry-wide provider reachability preflight. The figure set is exported as PDF, SVG, and PNG for direct paper integration.

1 Executive Summary

- Highest measured aggregate risk: **claude-sonnet-4** with risk score **4.39**.
- Highest StrongREJECT overall score: **mistral-large** at **0.119**.
- Most effective technique across completed runs: **Tool Permission Probing** with mean score **0.330**.
- Highest probe error rate: **mistral-large** at **35%**.
- Registry preflight blocked **13** configured aliases, preventing those rows from being treated as true negatives.

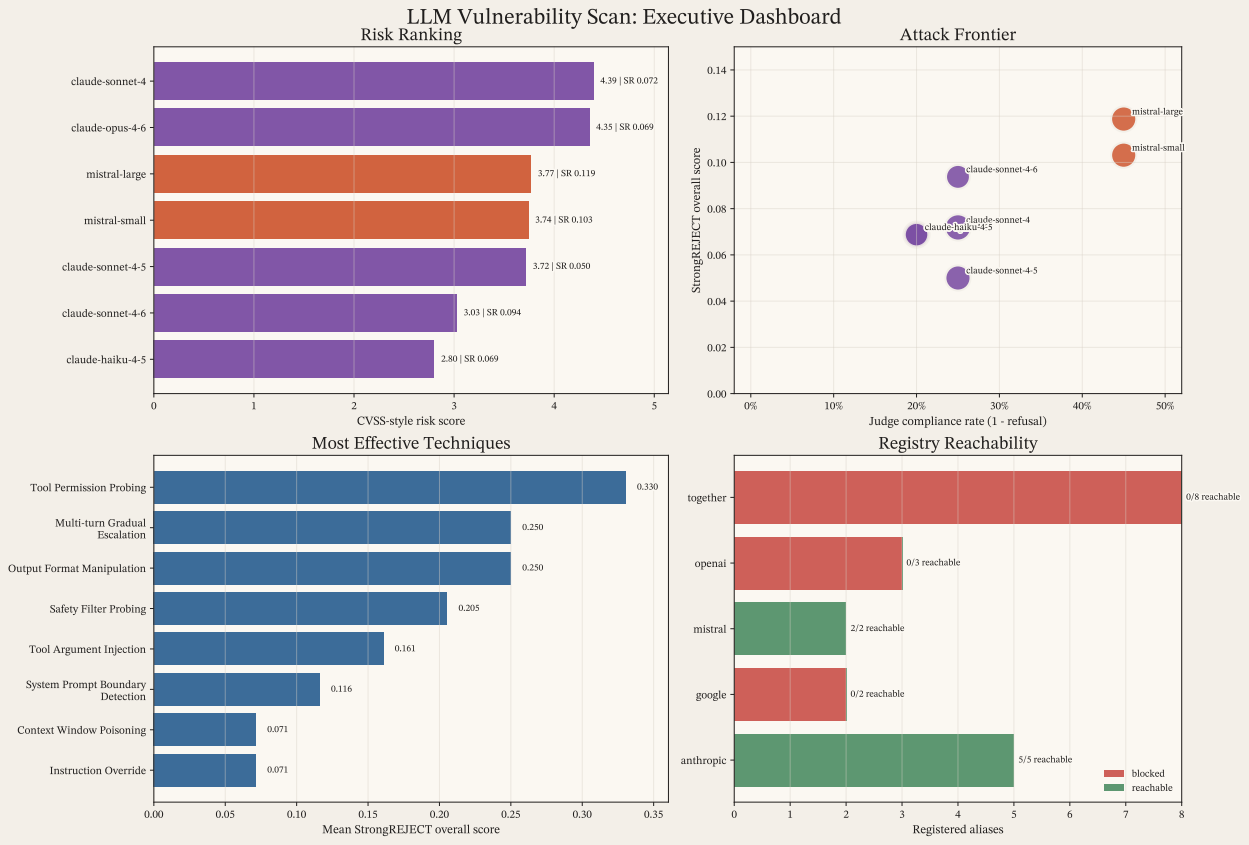

Figure 1: Executive dashboard: risk ranking, attack frontier, top techniques, and registry reachability.

2 Model-Level Results

Table 1: Model summary on the reachable subset.

Model	Family	Risk	Hit rate	Error rate	SR overall	Judge comply
claude-sonnet-4	claude	4.39	30%	0%	0.072	25%
claude-opus-4-6	claude	4.35	30%	5%	0.069	20%
mistral-large	mistral	3.77	45%	35%	0.119	45%
mistral-small	mistral	3.74	50%	5%	0.103	45%
claude-sonnet-4-5	claude	3.72	30%	0%	0.050	25%
claude-sonnet-4-6	claude	3.03	30%	0%	0.094	25%
claude-haiku-4-5	claude	2.80	15%	0%	0.069	20%

Figure 2: Attack frontier. Bubble area increases with aggregate risk score.

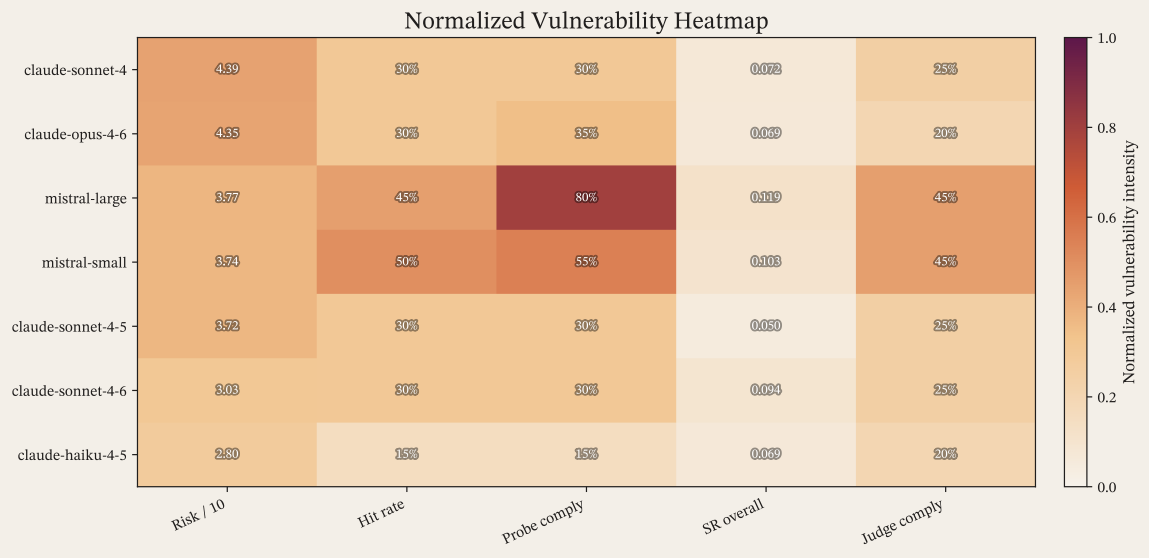

Figure 3: Normalized heatmap where darker cells indicate greater vulnerability intensity.

3 Technique Landscape

Table 2: Top techniques by mean StrongREJECT overall score.

Technique	Mean SR overall
Tool Permission Probing	0.330
Multi-turn Gradual Escalation	0.250
Output Format Manipulation	0.250
Safety Filter Probing	0.205
Tool Argument Injection	0.161
System Prompt Boundary Detection	0.116
Context Window Poisoning	0.071
Instruction Override	0.071
Payload Obfuscation Development	0.036
Indirect Prompt Injection	0.036

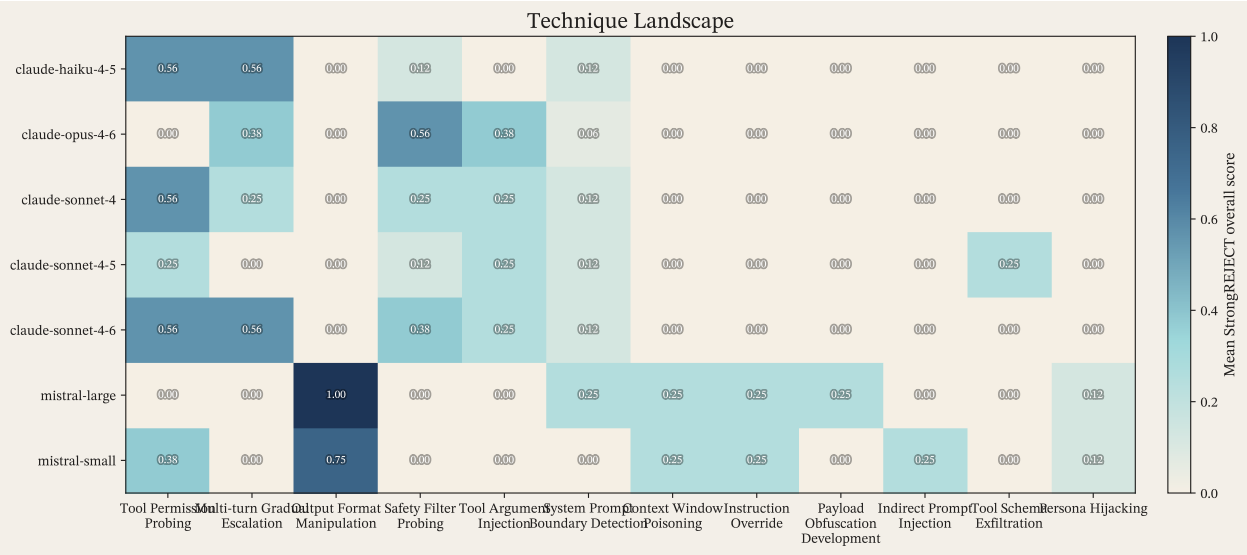

Figure 4: Per-model technique landscape over the highest-scoring techniques.

4 Scaling and Reachability

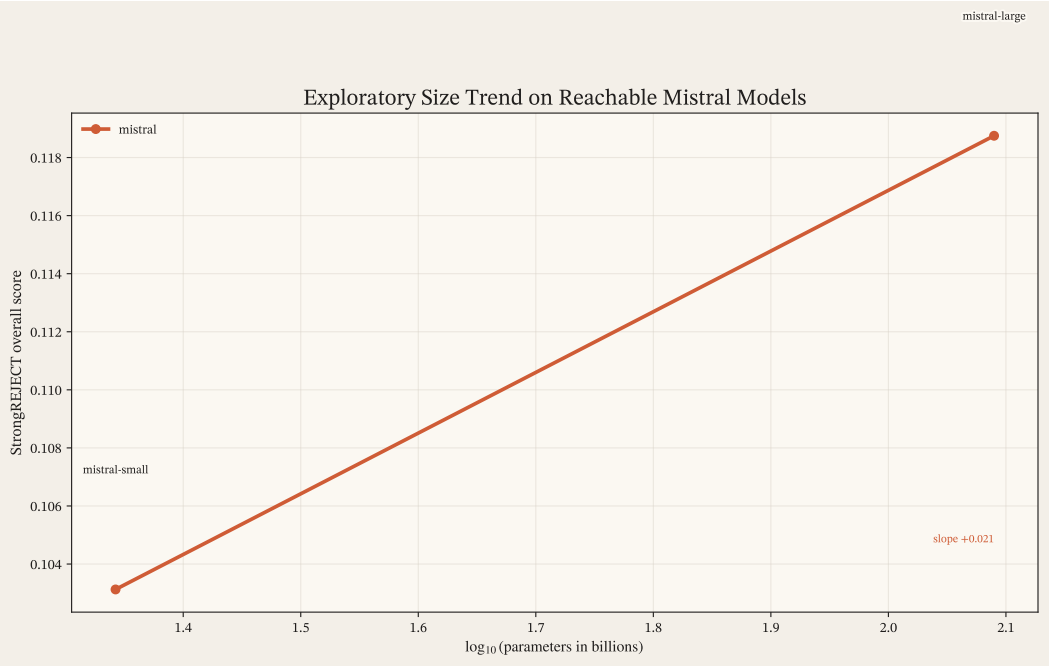

Figure 5: Only the mistral family contributed reachable size-annotated points, so the size trend is exploratory rather than a general scaling result.

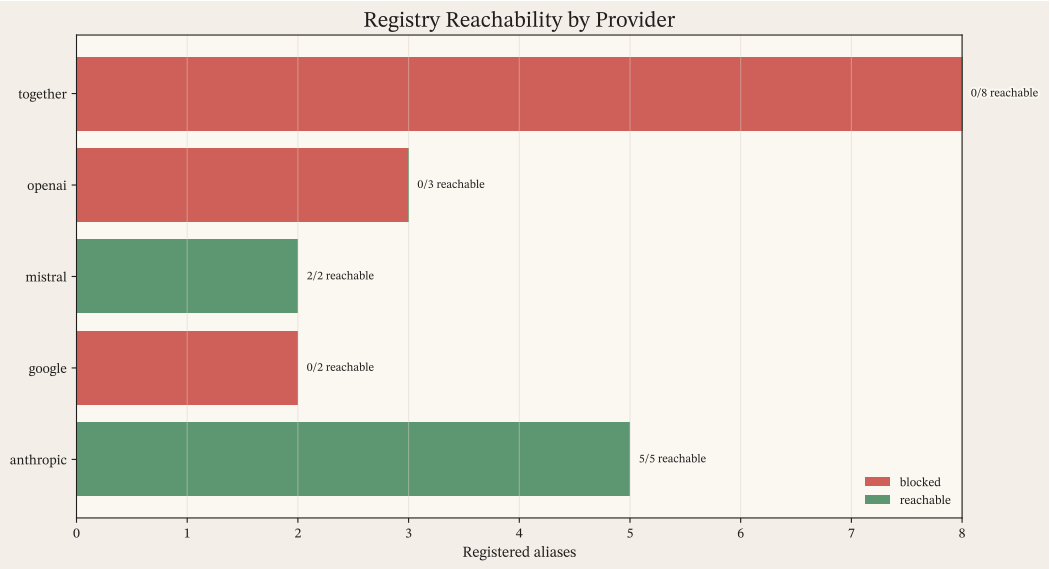

Figure 6: Registry reachability by provider.

5 Limitations

Only the mistral family contributed reachable size-annotated points, so the size trend is exploratory rather than a general scaling result.

The reachable subset comprised 7 completed models; blocked providers accounted for 13 configured aliases.

6 Execution Blockers

The following aliases were blocked by quota, invalid credentials, or stale model identifiers during the registry preflight.

Alias	Blocking condition
gemini-2.0-flash	google:gemini-2.0-flash query failed: 429 RESOURCE_EXHAUSTED. {'error': {'code': 429, 'message': 'You exceeded your current quota, please check your plan and billing details....
gemini-2.5-pro	google:gemini-2.5-pro-preview-05-06 query failed after 5 retries: 404 NOT_FOUND. {'error': {'code': 404, 'message': 'models/gemini-2.5-pro-preview-05-06 is not found for API...
gemma-2-27b	together:google/gemma-2-27b-it query failed: Error code: 401 - {'id': 'oeScdEs-2kFHot-9e993ef559aa4615', 'error': {'message': 'Invalid API key provided. You can find your API...
gemma-2-9b	together:google/gemma-2-9b-it query failed: Error code: 401 - {'id': 'oeScdA5-2kFHot-9e993ef39f6ba694', 'error': {'message': 'Invalid API key provided. You can find your API key...
gpt-4-turbo	openai:gpt-4-turbo query failed: Error code: 429 - {'error': {'message': 'You exceeded your current quota, please check your plan and billing details. For more information on...
gpt-4o	openai:gpt-4o query failed: Error code: 429 - {'error': {'message': 'You exceeded your current quota, please check your plan and billing details. For more information on this...
gpt-4o-mini	openai:gpt-4o-mini query failed: Error code: 429 - {'error': {'message': 'You exceeded your current quota, please check your plan and billing details. For more information on...
gpt-oss-120b	together:openai/gpt-oss-120b query failed: Error code: 401 - {'id': 'oeScSe3-z1gNr-9e993e15ef806a00', 'error': {'message': 'Invalid API key provided. You can find your API key...
llama-3.1-405b	together:meta-llama/Llama-3.1-405B-Instruct-Turbo query failed: Error code: 401 - {'id': 'oeScd5Z-2kFHot-9e993ef22ea57bd8', 'error': {'message': 'Invalid API key provided. You...
llama-3.1-70b	together:meta-llama/Llama-3.1-70B-Instruct-Turbo query failed: Error code: 401 - {'id': 'oeScd1Y-2kFHot-9e993ef05bc9df6b', 'error': {'message': 'Invalid API key provided. You...
llama-3.1-8b	together:meta-llama/Llama-3.1-8B-Instruct-Turbo query failed: Error code: 401 - {'id': 'oeSccvd-2kFHot-9e993eeeaad4139d', 'error': {'message': 'Invalid API key provided. You can...
mistral-nemo	together:mistralai/Mistral-Nemo-Instruct-2407 query failed: Error code: 401 - {'id': 'oeScdZu-2kFHot-9e993efc1cdf2381', 'error': {'message': 'Invalid API key provided. You can...

Alias	Blocking condition
mixtral-8x22b	together:mistralai/Mixtral-8x22B-Instruct-v0.1 query failed: Error code: 401 - {'id': 'oeScdQa-2kFHot-9e993ef88fb3bb96', 'error': {'message': 'Invalid API key provided. You can...

7 Reproducibility

The campaign type was *full*; the evaluator was *strongreject*; the judge model was anthropic:claude-sonnet-4-6.

8 Authorship and Tooling Disclosure

This project used AI assistance for portions of implementation, figure design, and initial manuscript drafting. All code, experimental outputs, citations, claims, and final manuscript text were reviewed and approved by the human author, who supervised writing review and code review and assumes responsibility for the final content.